

The Tasks:

Task 1: Build the basic topology of the network and get it running in Cisco Packet Tracer.

- 1.1.** This includes basic router and switch setup and configuration.
- 1.2.** Any DHCP/SLAAC services that are needed.
- 1.3.** Set up any VLANs (should you wish to use them) and routing protocols as necessary to give you full connectivity across the network before security features are added.

Task 2: Apply the required internal security features to that network and then submit it as a second simulator file.

- 2.1.** This should include everything from Task 1 and:
- 2.2.** Basic device security.
- 2.3.** The setup and configuration of firewalls, which could be packet filters or stateful zone-based ones as appropriate, which can be router or security appliance based. You will need to add these to the topology where you feel appropriate as this has not been done yet and adjust your IP scheme to fit too.
- 2.4.** The use of NAT and PAT where and if needed. Tell us why it is needed and why you placed it where you did if you use it. Alternatively, tell us why you are not using it.
- 2.5.** The securing of routing protocol advertisements with suitable passwords, to prevent unauthorised disclosure of routing information.

Task 3: Apply the required external security features between the two sites in a copy of your Task 2 simulation, and submit that as a third simulator file.

- 3.1.** Implementing GRE tunnels where needed.
- 3.2.** Implementing site-to-site VPNs where needed.

Task 4: Provide a commentary describing how the security features were added and why they are there.

- 4.1.** This may include why a feature has not been used in your solution.
- 4.2.** Your choice of IP version and why.
- 4.3.** Details of the implementation of all the security features.
- 4.4.** Reasoning behind the use of firewall types and/or NAT/PAT deployment.

4.5. Reasoning behind the choice of VPN parameters and use of GRE tunnels.

Task 5: Provide a sample of five rules that you would include in the security policy document for the company to help enforce stable security across this network.

Network Diagram

